IPsec, SSL VPN, and Cisco Configuration

IE3122: Network Security | Week 9

What's actually happening inside the two dominant VPN technologies

Today's Agenda

- IPsec: architecture, components, and modes
- SSL VPN: architecture, client access modes, and its modular cryptographic building blocks
- IPsec versus SSL VPN, compared directly
- Configuring tunnels and VPNs on Cisco devices: concept and template, for GRE, IPsec, and GRE over IPsec

Learning Outcomes for Today

LO3

– Analyse security requirements and recommend appropriate technologies.

LO4

– Evaluate different security technologies, compare them, and identify appropriate use cases.

LO5

– Configure security technologies to satisfy specific requirements.

IPsec VPN

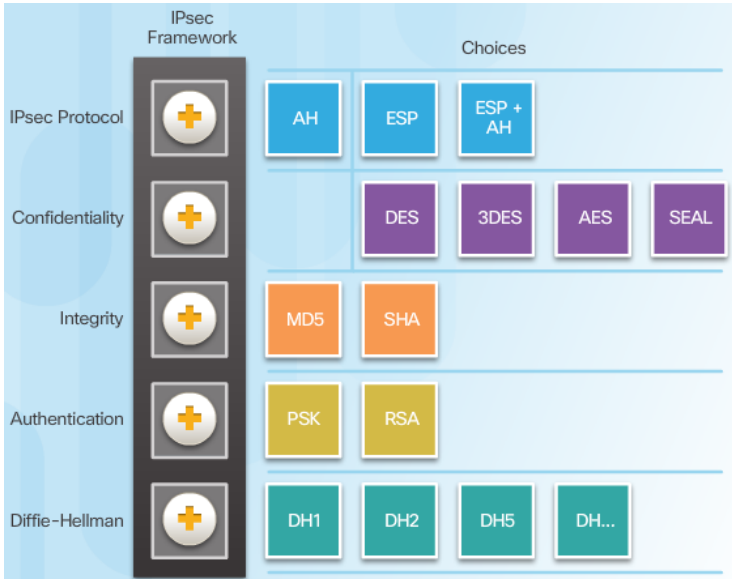
Securing the network layer itself

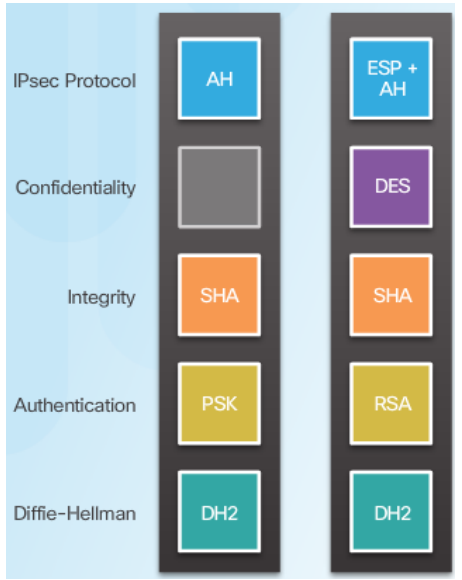
IPsec: Overview

- IPsec (Internet Protocol Security) operates at the network layer, securing IP packets themselves, regardless of which application generated them.
- It is not one fixed protocol but a modular framework: it deliberately does not mandate which specific algorithm you use for any given job, only which job needs doing.
- Because it protects at the IP layer, IPsec secures everything above it automatically. No individual application needs to be aware it's even running.

IPsec as a Modular Framework

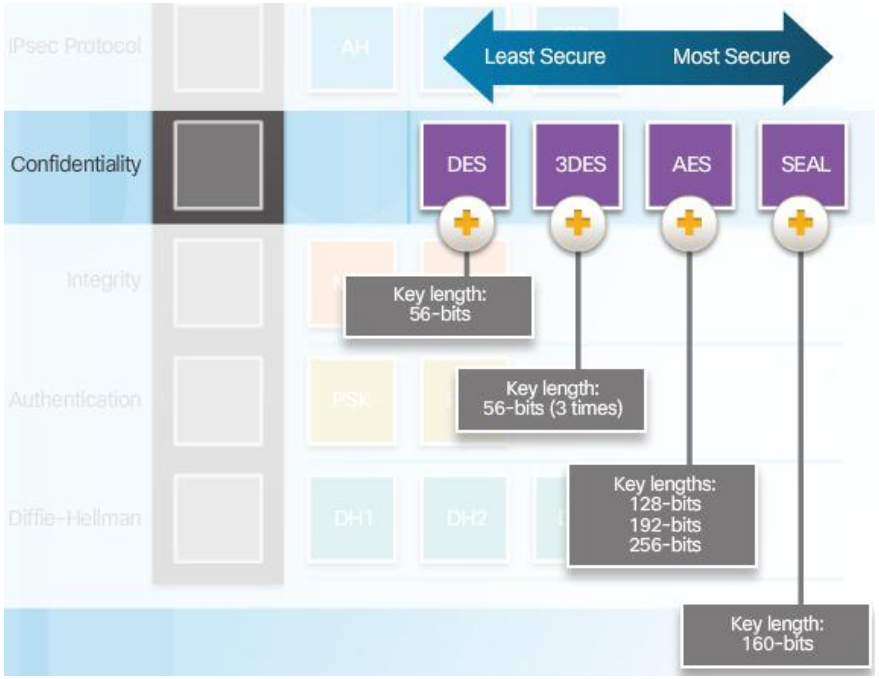
- IPsec defines five separate functions, each with a menu of interchangeable choices rather than one mandatory answer. The first row matters most: AH authenticates only, with no encryption; ESP adds real confidentiality, which is why ESP or ESP+AH is used far more often.
- An actual configuration picks exactly one choice per row. The two examples on the right are both valid, built from that same menu.

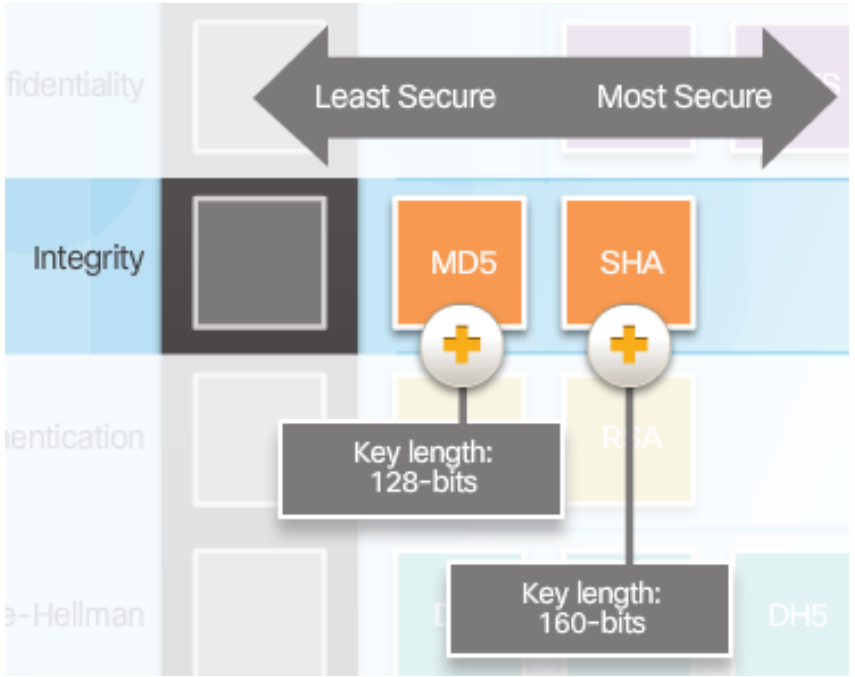




Choosing Confidentiality and Integrity Algorithms

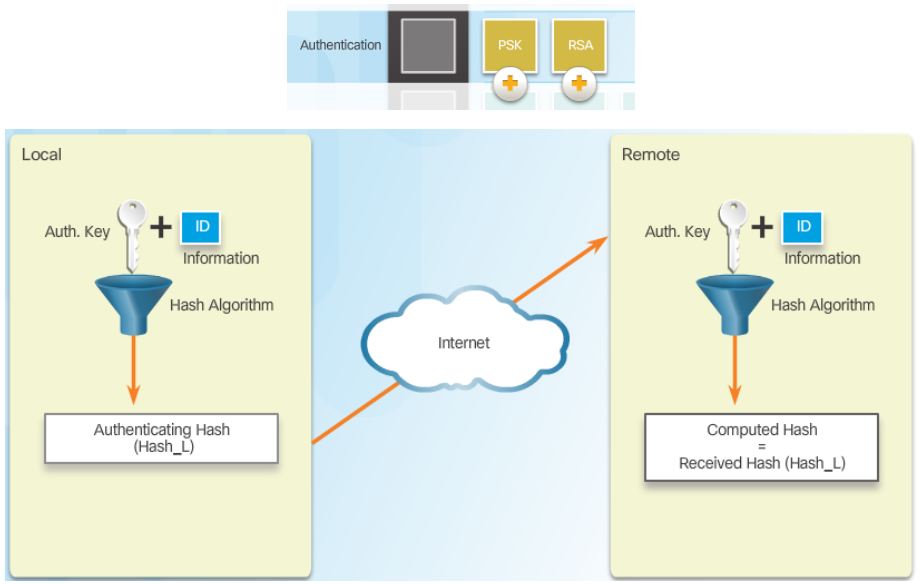
– Confidentiality and integrity are two separate rows in the same framework, each with its own menu, and each option carries a genuinely different security strength.





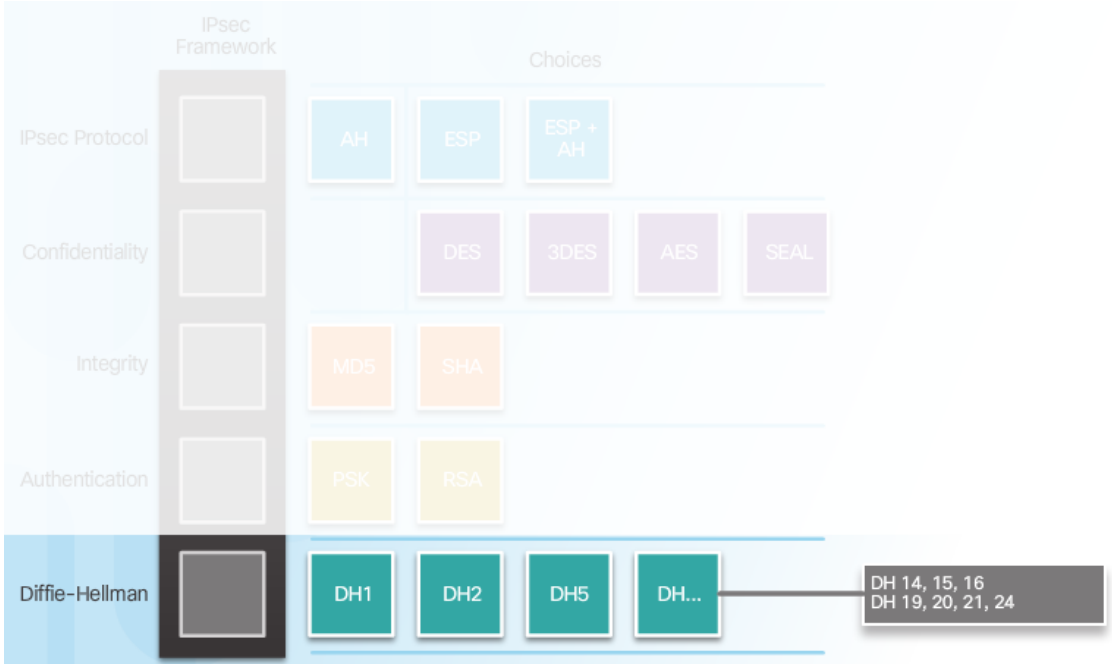
Peer Authentication Methods

- The Authentication row offers two choices: PSK (pre-shared key, the same secret configured manually on both peers) or RSA (certificate-based, using public-key cryptography instead).
- Either way, the mechanism is the same shape: each peer hashes its own authentication key together with its identity, and the two sides compare hashes to confirm they agree, without ever sending the key itself across the network.



Diffie-Hellman Group Choices

- The Diffie-Hellman row picks which mathematical group the key exchange itself uses. Higher-numbered groups generally mean larger, stronger keys.
- This is a genuinely live consideration, not just historical: DH14 is the realistic minimum for production networks today, but older or simulated equipment doesn't always support it.



IPsec: IKE's Two Phases

- IKE Phase 1 establishes a secure, authenticated channel between the two peers themselves, called the ISAKMP SA. This is the peers proving who they are to each other and agreeing on how to protect their own negotiation traffic.
- IKE Phase 2 then uses that secure channel to negotiate the actual IPsec Security Associations that will protect real data traffic, the specific encryption and authentication parameters applied to the traffic itself.
- Splitting negotiation into two phases means the peers only need to re-authenticate each other occasionally, in Phase 1, while Phase 2 can refresh the data-protection keys far more frequently without repeating full peer authentication every time.

IPsec: Transport Mode vs Tunnel Mode

- Transport mode encrypts only the payload of the original IP packet, leaving the original IP header intact. This is typically used for direct host-to-host protection, where both ends are the actual communicating parties.
- Tunnel mode encrypts the entire original IP packet, including its header, and wraps it inside a new IP packet with new addressing. This is what site-to-site VPNs use, since it hides the original internal addressing entirely from the public network in between.
- The choice of mode is independent of whether you're using AH or ESP; either component can run in either mode, though ESP in tunnel mode is by far the most common real-world combination.

IPsec: Strengths and Limitations

- Strengths: operates at the network layer, so it's completely application-transparent, protecting all IP traffic without needing any individual application to support it. It's a strong, mature, widely interoperable standard with robust authentication options.
- Limitations: configuration genuinely is complex, involving several distinct components, an ISAKMP policy, a transform set, and traffic-matching rules, that all have to agree correctly. AH in particular struggles to cross NAT devices, since NAT modifies the very IP header AH is protecting, breaking its integrity check; ESP needs a NAT Traversal extension to reliably cross NAT too. It's also generally less naturally suited to simple browser-based remote access than SSL VPN.

SSL VPN

Built from interchangeable cryptographic building blocks

SSL VPN: Overview

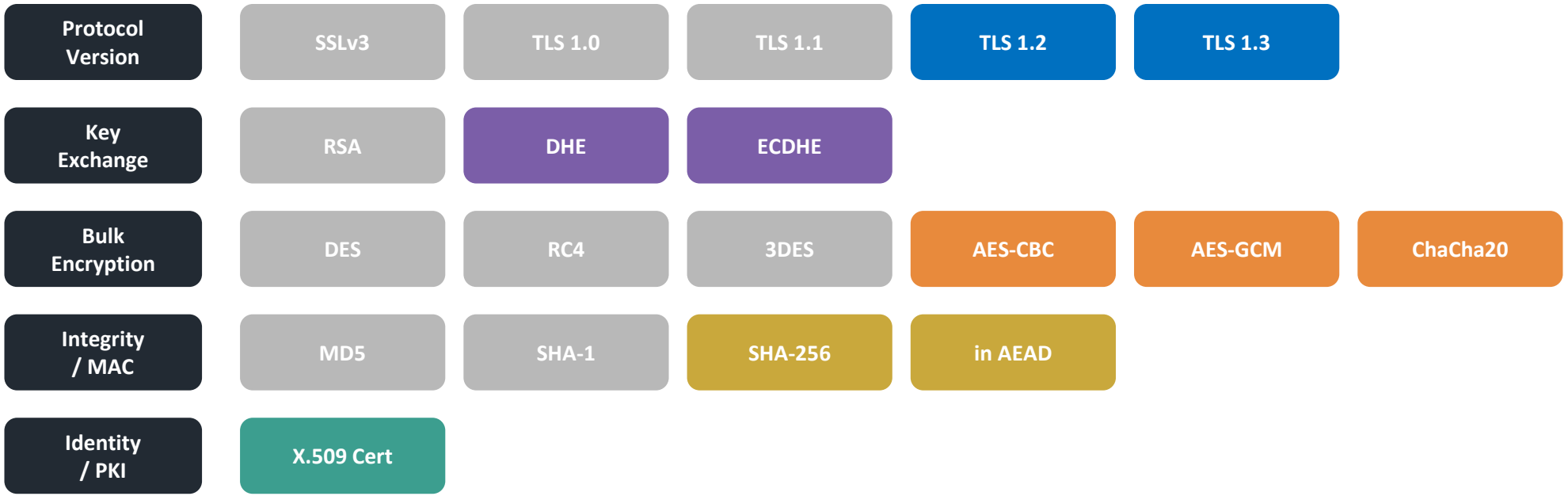
- SSL VPN uses the same TLS/SSL protocol that secures ordinary HTTPS web traffic, rather than a dedicated network-layer protocol suite like IPsec.
- Because it rides on the same protocol as HTTPS, it often needs nothing more than a standard web browser, or at most a lightweight client, rather than dedicated VPN client software.
- It also tends to traverse firewalls and NAT more easily than IPsec in practice, simply because almost every corporate firewall already permits outbound HTTPS on port 443, so SSL VPN traffic blends straight into traffic that was already allowed.

SSL VPN Has Interchangeable Building Blocks Too

- IPsec is the technology most commonly held up as the modular one, precisely because of the explicit, named framework you just saw. SSL/TLS is built the same underlying way, just negotiated differently: as a single named cipher suite chosen from a list, rather than assembled row by row.
- SSL/TLS is not one fixed algorithm. It's built from several independent functional building blocks, and the two peers negotiate which specific option fills each block at the start of every session.
- This matters directly for security: an SSL VPN configured to allow older, weaker options in any one of these blocks can be seriously weakened, even if every other block is configured well.
- The next slide breaks down exactly which functional block does what, and which real protocols and algorithms have historically filled each one.

SSL VPN's Functional Blocks, Visually

– The same idea as IPsec's framework, laid out the same way: one row per function, each with a menu of choices. Greyed-out options are deprecated and should never be selected today.



Building Two Example Cipher Suites

– Pick one option from each row and you get a real, nameable cipher suite. These two are both valid TLS configurations, but they are not remotely equivalent in strength.

	Weaker (avoid)	Stronger (prefer)
Protocol Version	TLS 1.2	TLS 1.3
Key Exchange	RSA	ECDHE
Bulk Encryption	AES-CBC	AES-GCM
Integrity / MAC	SHA-1	built into cipher

SSL VPN's Building Blocks, Function by Function

Function	What it does	Protocols/algorithms that can fill it
Protocol version	Which version of TLS governs the whole session	SSLv3, TLS 1.0, TLS 1.1 (all deprecated); TLS 1.2 (widely used); TLS 1.3 (current)
Key exchange	How the two sides agree on a shared secret	RSA (no forward secrecy); DHE; ECDHE (modern, provides forward secrecy)
Bulk encryption	The cipher actually protecting the data	DES, RC4 (both insecure); 3DES (deprecated); AES-CBC; AES-GCM; ChaCha20
Integrity/MAC	Detecting if data was tampered with	MD5, SHA-1 (both deprecated); SHA-256/384; built into AEAD ciphers like AES-GCM
Identity/PKI	Verifying the server's, and optionally the client's, identity	X.509 certificates, validated through a Certificate Authority chain

SSL VPN: Client Access Modes

- The cryptography isn't the only modular part of SSL VPN. How a client actually connects is also independently selectable.
- Clientless (portal) mode: an ordinary web browser only, reaching a limited set of internal web applications through a secure portal.
- A third mode, thin client (port-forwarding), sits between the two shown below: a small browser applet extends access to specific TCP applications without a full install.
- Full tunnel client mode: a complete agent, such as Cisco AnyConnect, providing full network-layer access, using TLS as the underlying transport.

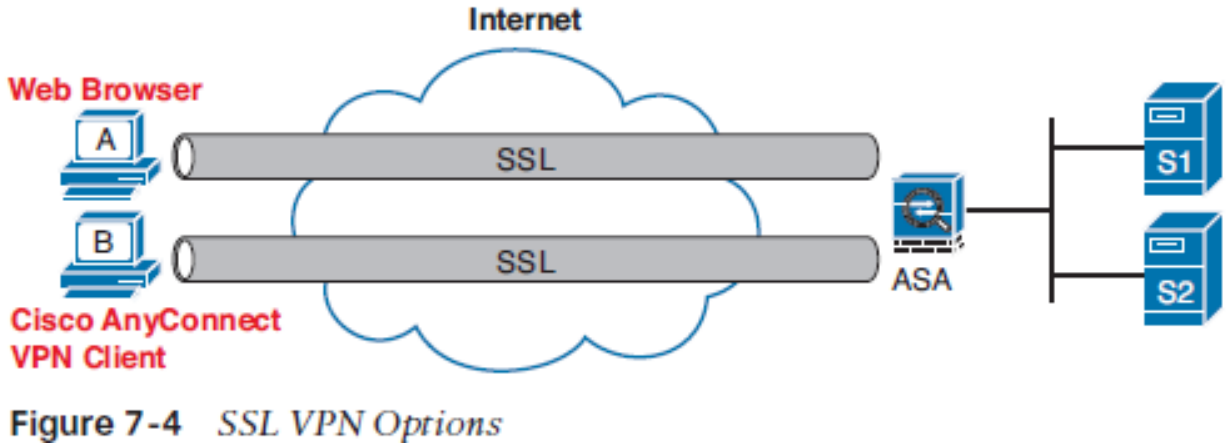


Figure 7-4 *SSL VPN Options*

Figure 7-4, SSL VPN Options: the clientless browser path (A) and the AnyConnect full-tunnel path (B), both terminating at the same ASA.

IPsec vs SSL VPN

Two different answers to the same underlying problem

IPsec vs SSL VPN: Direct Comparison

Aspect	IPsec VPN	SSL VPN
Layer of operation	Network layer (Layer 3)	Built on TLS, above the transport layer
Client requirement	Often needs dedicated client software	Often just a browser, in clientless mode
Traffic scope	Secures all IP traffic transparently	Typically specific applications or resources, though full-tunnel mode approaches IPsec's scope
NAT traversal	Can struggle, especially AH; needs NAT-T	Generally NAT-friendly, uses standard HTTPS
Access granularity	Network-level, largely all-or-nothing per tunnel	Can be very granular, down to specific applications or URLs
Typical use case	Site-to-site links, full-network remote access	Individual remote access, especially browser-based

Configuring VPNs and Tunnels on Cisco Devices

Concept alongside the actual command templates

GRE Tunnel: Concept

- A GRE tunnel is configured as a logical Tunnel interface, entirely separate from any physical interface, with its own IP addressing for the virtual point-to-point link it creates.
- It needs a source, the local physical interface or address traffic will actually leave from, and a destination, the remote peer's real, public address.
- Remember from earlier: this alone provides no encryption whatsoever. GRE just gets traffic there, in the clear.

```
interface Tunnel0
 ip address 10.10.10.1 255.255.255.252
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.10
```

IPsec Site-to-Site: Step 1, the IKE Phase 1 Policy

– The first step defines how the two peers will authenticate each other and protect their own negotiation traffic, the ISAKMP policy from Phase 1 earlier in this deck.

```
crypto isakmp policy 10
 encryption aes
 hash sha256
 authentication pre-share
 group 14
 lifetime 86400
exit
crypto isakmp key MySharedKey address 203.0.113.10
```

IPsec Site-to-Site: Step 2, the Transform Set

– The transform set defines the Phase 2 parameters, the actual encryption and authentication that will protect real data traffic once negotiation is complete.

```
crypto ipsec transform-set MYSET esp-aes esp-sha256-hmac
```

IPsec Site-to-Site: Step 3, Defining Interesting Traffic

– An ACL identifies exactly which traffic should actually be sent through the VPN, commonly called interesting traffic. Only traffic matching this ACL gets encrypted; everything else follows normal routing.

```
access-list 100 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
```

IPsec Site-to-Site: Step 4, the Crypto Map

– The crypto map ties everything together: which peer to protect traffic toward, which transform set to use, and which ACL defines interesting traffic, then applies the whole thing to the outbound interface.

```
crypto map MYMAP 10 ipsec-isakmp
 set peer 203.0.113.10
 set transform-set MYSET
 match address 100
exit
interface GigabitEthernet0/0
 crypto map MYMAP
```

GRE Over IPsec: Bringing Both Together

- Configure the GRE tunnel exactly as shown earlier, giving you a working but unencrypted logical link.
- Then configure IPsec exactly as in the previous four steps, with one key difference: the interesting-traffic ACL should match GRE traffic itself, protocol 47, between the tunnel endpoints, rather than the traffic riding inside the tunnel.
- The result carries the best of both: GRE's flexibility, including support for dynamic routing protocols and non-IP traffic across the tunnel, protected by IPsec's actual encryption. This is a genuinely common real-world design, not just a teaching exercise.

Summary

- IPsec secures the network layer itself using a suite of cooperating protocols, AH for integrity alone, ESP for encryption, and IKE to negotiate it all in two phases.
- SSL VPN rides on the same TLS that secures HTTPS, and is genuinely modular: the protocol version, key exchange, bulk cipher, and integrity check are all independently negotiated, and the client access mode is a separate choice again.
- Neither is simply better than the other. IPsec suits full-network, always-on connections; SSL VPN suits individual, often browser-based remote access.
- GRE, IPsec, and GRE over IPsec each solve a different problem, plain transport, plain security, or both together, and the configuration for each builds directly on the last.

This Week's Lab & Tutorial

- Lab: to be planned in a follow-up discussion, once we've confirmed which of today's configurations are genuinely supported and testable in Packet Tracer.
- Tutorial worksheet: study-notes-style questions covering IPsec, SSL VPN's modular building blocks, and Cisco VPN configuration. Use it to prep for the mid-semester test.

References

- Santos & Stuppi, CCNA Security 210-260 Official Cert Guide, 1st Ed., Cisco Press.
- Kaufman, Perlman, Speciner & Perlner, Network Security: Private Communication in a Public World, 3rd Ed., Prentice Hall.
- Cisco Networking Academy, CCNA Security curriculum (IPsec and SSL VPN chapters).